Charlotte Ward

Assistant Professor
Department of Astronomy and Astrophysics
The Pennsylvania State University
Davey Laboratory, 251 Pollock Road, University Park, PA 16802

https://charlotteaward.github.io
cvw5890@psu.edu

Education

Ph.D., Astronomy, University of Maryland at College Park — 2022
Thesis: *Tracing the formation and merger-driven growth of massive black holes with the Zwicky Transient Facility*

M.Sc., Astronomy, University of Maryland at College Park — 2019

B.Sc. (Adv) (Hons), Physics & Mathematics, University of Sydney — 2016
Thesis: *Machine learning techniques for discovery of Fast Radio Bursts with the Parkes Radio Telescope*

Professional Employment and Internships

Assistant Professor, The Pennsylvania State University	2025 –
Postdoctoral Research Associate, Princeton University	2022 – 2025
HEP Center for Computational Excellence Graduate Summer Internship, Lawrence Berkeley National Laboratory	2019
School of Physics Laboratory Tutor, University of Sydney	2017
Summer Scholar, Centre for Astronomy and Space Science, CSIRO	2014, 2015
Nanjing Research Exchange Visiting Scholar, Nanjing University	2015
Faculty of Science Information Officer, University of Sydney	2014 – 2015
Undergraduate research student, Quantum Control Laboratory, University of Sydney	2015
Undergraduate research student, Institute of Medical Physics, University of Sydney	2014

Research interests

Multi-resolution image modeling techniques for joint analysis of ground and space-based surveys; multi-wavelength analysis of astrophysical transients; dwarf galaxy AGN; changing-look AGN; tidal disruption events; strongly lensed QSOs and SNe; SMBH binaries; machine learning for transient classification and light curve prediction; large-scale pipelines for wide-field time-domain surveys.

Grants and Observing Proposals (Principal Investigator)

Revealing the hidden population of wandering massive black holes with Rubin Observatory Early Data (LSST Discovery Alliance, $117K awarded) — 2026

A Deep Dive into Massive Black Holes in the Era of Rubin: From Optical Variability to Multi-Messenger Signatures (Institute for Gravitation and the Cosmos, Penn State, $7K awarded) — 2026

Exploring the physical origin of compact millimeter emission in radio-quiet AGN with Swift X-ray monitoring (NASA Swift, $8K awarded) — 2025

Mercedes Richards Career Development Professorship (Penn State, $60K awarded) 2025
A Scarlet2 framework for characterizing transients and their host galaxies
(LSST Discovery Alliance LINCC Frameworks Incubator Program, $22.5K awarded) 2025
Institute for Computational and Data Sciences Rising Researcher Collaborations Seed Grant (Penn State, $18K awarded) 2025
Unveiling disk formation and evolution in rebrightening tidal disruption events (WIYN) 2025
Understanding variable radio emission in changing-look AGN (VLA) 2025
Joint modeling of imaging data from LSST and complementary surveys to maximize early transient science (LSST Discovery Alliance Small Grants, $8K awarded) 2024
Understanding young radio jets in changing-look AGN with ATCA (ATCA) 2024
Understanding episodic SMBH accretion triggering with changing-look AGN (Magellan) 2023
Investigating 10 Candidates for Gravitational Wave Recoil from an SMBH merger (Keck, NASA allocation, $15K awarded) 2023
Confirming the presence of AGN for the variability-selected IMBH candidates from ZTF and WISE (Magellan) 2023
CoI: Various HST, Chandra, VLA and ATCA proposals 2023 - 2025

Research Advising

Graduate students

Pei-Jun Huang (1st year): Roman grism deblending via joint survey forward modeling 2026 –
Sophie Bergstrom (1st year): Superluminous supernova progenitors 2026 –
Kendall Sippy (2nd year, NSF GRFP awarded 2026): Non-nuclear MBHs in Rubin/Euclid 2025 –
Miranda Zak (3rd year, PSGC Fellow): Millimeter/X-ray studies of radio-quiet AGN 2024 –
Maddy Reich (3rd year): Host galaxy studies with Rubin/Euclid, changing-look AGN 2026 –
Chloe Klare (6th year, PSGC Fellow): Young radio jets in AGN 2024 –

Undergraduate students

Brady Barth (Penn State astro): Junior research project 2026 –
Dheer Patel (Penn State astro): Junior research project 2026 –
Alex Garner (Penn State astro): Junior research project 2026 –
Samantha Baugh (Penn State astro): 2 x Junior research project 2026 –
Russell Dunlap (Penn State astro): Sophomore and junior research projects 2026 –
Grace Davis (Penn State astro → UHawaii REU): 2 x Sophomore research project 2025 –
Naren Arun (Penn State astro): 2 x Junior research project 2025 –
Adam Papazoglou (Penn State astro, primary advisor Kendall Sippy): Junior research project 2026 –
Veena Krishnaraj (Princeton astro → UMD PhD): Junior Thesis, Senior Thesis → paper in prep 2025 –
Sufia Birmingham (Princeton astro → Oxford PhD):

2 x Undergraduate Summer Research Program, Junior Thesis → AAS Chambliss award
1 submitted first-author publication, 3 co-authored publications 2022-2026

Hy Truong (Princeton astro → MSc student SDSU): Junior Thesis, Senior Thesis 2023-2025

Anavi Uppal (Yale astro—> PhD student and NSF fellow at UCSC):
1 accepted first-author publication, AAS Chambliss award 2023

Sophie Chen (Princeton engineering): Senior Thesis 2023

Abigail García-Pérez (GRAD-MAP student → PhD student University of Turin):
Winter Workshop project → 1 co-authored publication 2019

Immaculate Oyoo (GRAD-MAP student):
Winter Workshop project → 1 co-authored publication 2019

PhD thesis committee membership

Prince Yadav (UIUC): ML methods for source deblending in Rubin 2026 –
Aditi Gangadharan (Penn State): X-ray diffraction gratings 2026 –
Bojun Zhang (Penn State): Radio AGN 2026 –
Thomas Gaudin (Penn State): X-ray binaries 2026 –
Tristen Sextro (Penn State): X-ray diffraction gratings 2026 –
Jean Feeser (Penn State): Stellar variability, SNIa calibration 2026 –
Kyle Neumann (Penn State): ML techniques for X-ray/gamma-ray source classification 2025 –
Mary Ogborn (Penn State): Accretion disk simulations for reverberation mapping 2025 –
Matt Sampson (Princeton): Latent ODEs for transient light curve prediction 2025 –

Professional Service and Leadership

Referee for MNRAS, ApJ, Nature, Nature Astronomy, A&A, JOSS 2022 - ongoing
Rubin Users Committee, Committee Member 2026 - ongoing
Discussion Panelist, LSST Discovery Alliance LINCC Frameworks Cycle 11 Review 2026
Discussion Panelist and Deputy Chair, NASA XRISM Cycle 3 Review 2026
Discussion Panelist, LSST Discovery Alliance LINCC Frameworks Cycle 10 Review 2026
Voting Member, Hobby Eberly Users Committee 2026 - ongoing
Discussion Panelist, Hubble Space Telescope Cycle Cycle 33 Review 2025
Discussion Panelist, NSF Division of Astronomical Sciences 2025
LSST Builder Status, For 2 years FTE of direct observatory efforts 2025
External proposal referee, Hubble Space Telescope DDT proposals 2025, 2026
Discussion Panelist, Hubble Space Telescope Cycle 32 Review 2024
Invited Speaker, VLASS Epoch 4 Review Panel 2024
Member, LS4 Black Holes WG 2025 - ongoing
Member, LSST DESC Pixels to Light Curves WG 2025 - ongoing
Member, Rubin/Euclid Derived Data Products WG 2024 - ongoing
Member, BASS survey collaboration 2024 - ongoing

Member, VLASS Survey Science WG	2024 - 2025
Member, LSST AGN Science WG	2024 - ongoing
Astrocoffee (arXiv journal club) host, Princeton University	2023 - 2025
Member, ZTF Black Holes WG	2017 - 2022
External proposal referee, CanTAC Gemini	2022
Co-coordinator of the Department of Astronomy Journal Club, University of Maryland	2019-2021

Teaching: Classroom and Workshops

ASTRO 480 Galaxies and Cosmology, Penn State	2026
ASTRO 589 Graduate Seminar: Survey Science, Penn State	2025
Lecturer, Undergraduate Summer Research Program seminars, Princeton University	2024, 2025
Instructor, GROWTH time-domain astronomy school, virtual.	2020
Teaching Assistant, 'Special Problems in Astronomy: Big Data', University of Maryland	2018
GRAD-MAP Python Bootcamp developer, University of Maryland	2018
Instructor, 3rd year computational physics lab, University of Sydney	2017
Instructor, 2nd year experimental physics lab, University of Sydney	2017
Grok Learning interactive programming course developer, University of Sydney	2017

EDI and Public Outreach

In-Service Workshops in Astronomy high school teacher training program: speaker, Penn State	2026
AstroFest, State College: speaker, volunteer, Penn State	2026
Society of Physics Students: guest speaker, Penn State	2026
Astronight, State College: speaker, volunteer, Penn State	2025
Astronomy on Tap, State College: speaker, Penn State	2025
Peyton Observatory Public Observing: volunteer/ lead observer/co-coordinator , Princeton U.	2022 - 2025
EDI seminar series coordinator, Princeton University	2023 - 2025
Astronomy on Tap Trenton: co-coordinator/speaker, Princeton University	2023 - 2024
Postdoc - Grad Student Mentoring Program mentor, Princeton University	2023 - 2025
Solar Eclipse Festival: co-coordinator, Princeton University	2024
'Science Under the Stars' speaker, New Jersey State Museum & Planetarium 2024	
Co-lead of GRAD-MAP (Graduate Resources Advancing Diversity in Maryland Astronomy and Physics) program, University of Maryland	2020-2021
Astronomy outreach volunteer for UMD observatory, UMD Open House, Public Library STEM events, and Community College visits, University of Maryland	2018-2021
Pulse@Parkes High School Outreach Program volunteer, CSIRO	2014-2015

Honors, Awards and Press Coverage

Scialog Fellow, Research Corporation for Science Advancement	2026
Associated Press Interview about Roman Space Telescope (84 TV broadcasts)	2026

Mercedes Richards Career Development Professorship, The Pennsylvania State U.	2025
Equity Prize for Outreach, Department of Astrophysical Sciences, Princeton U.	2024
Martin and Beate Block Winter Award, Aspen Center for Physics	2023
ZTF Recoiling AGN paper featured in New Scientist	2021
Graduate Student Summer Research Fellowship, University of Maryland at College Park	2020
Graduate School Dean's Fellowship for Astronomy, University of Maryland at College Park	2017
Physics Foundation Scholarship No III, University of Sydney	2016
School of Physics Honours Scholarship, University of Sydney	2016
Faculty of Science Merit Scholarship, University of Sydney	2013-2015
School of Physics Smith Prize in Experimental Physics, University of Sydney	2013

Selected Colloquia, Seminars and Conference Talks

Invited Colloquium, KIPAC, Stanford University	Scheduled 2027
Invited Colloquium, University of Illinois, Urbana-Champaign	2026
Invited Colloquium, University of Wisconsin, Milwaukee	2026
Invited Seminar, Texas A&M University, College Station	2026
Contributed Talk, Schmidt Sciences Symposium	2026
Contributed Talk, Cornell/Syracuse Research Day, Cornell University	2026
Contributed Talk, Rubin Community Workshop, SLAC	2026
Contributed Talk, LSST Dark Energy Science Collaboration meeting, Boston University	2026
Invited Talk, Hot-wiring the Next Generation of Time-Domain Science, Caltech	2026
Contributed Talk, Multi-wavelength approach to the AGN structures, Kyoto University	2026
Invited Talk, Accretion History of AGN, Miami	2025
Invited Colloquium, Drexel University	2025
Invited Seminar, University of Kentucky	2025
Invited Talk, LINCC Tech Talks, LSST Discovery Alliance	2025
Contributed Talk, Strongly Lensed Transients from Rubin: Readiness Workshop	2025
Invited Seminar, Berkeley Theoretical Astrophysics Center	2025
Contributed Talk, Data-Driven Discovery in the Rubin Era, SLAC	2025
Contributed Talk, Transients in Space, STScI	2025
Contributed Talk, Aspen Winter Conference: The Era of Binary SMBHs	2025
Contributed Talk, AAS Winter Meeting	2025
Internal Talk, Bahcall Lunch, Princeton U./IAS	2025
Invited Talk, HEACOSS-2024 conference	2024
Invited Talk, SIfA Morning Tea, University of Sydney School of Physics	2024
Invited Seminar, Naval Research Laboratory Remote Sensing Division	2024
Contributed Talk, Rubin Community Workshop	2024
Contributed Talk, Tidal Disruption Events and Nuclear Transients conference	2024
Contributed Talk, BASS2024 Team Meeting	2024

Internal Talk, Thursday Lunch Talks, Princeton U.	2024
Invited Colloquium, The Pennsylvania State University, Department of Astronomy	2023
Contributed Talk, IMBHs: The Dawn of a Revolutionary Era conference	2023
Invited Seminar, ASKAP Variable And Slow Transients (VAST) team meeting	2023
Contributed Talk, Aspen Winter Conference: eXtreme Black Holes	2023
Invited Colloquium, Lawrence Berkeley National Laboratory, Computer Sciences Division	2022
Invited Seminar, Stanford KIPAC Tea Talk	2022
Invited Colloquium, Lawrence Berkeley National Laboratory, Physics Division	2022
Invited Colloquium, Australia Telescope National Facility, CSIRO	2022
Invited Seminar, Nanograv Meeting	2022
Contributed Talk, IMBHS: New Science from Stellar Evolution to Cosmology conference	2022
Contributed Talk, Aspen Winter Conference: Dynamical Formation of GW Sources	2022
Invited Seminar, Johns Hopkins University AGN Journal Club	2021
Invited Seminar, NASA Goddard AGN Seminar Series	2021
Contributed Talks, ZTF Team Meetings in Stockholm, Tel Aviv, and Pasadena	2018-2019

Selected Peer-Reviewed and Submitted Publications

51 total published, h-index=27, 6238 total citations.

*: graduate student †: undergraduate student +: postdoc

Ward, C., Koss, M. et al. 'BASS. LII. The Prevalence of Double-peaked Broad Lines at Low Accretion Rates among Hard X-Ray Selected Active Galactic Nuclei', **2025,** ApJ, 991, 596.

Ward, C., Melchior, P., Sampson, M.* et al. 'Disentangling transients and their host galaxies with Scarlet2: A framework to forward model multi-epoch imaging', **2025**, Astronomy and Computing, 51, 100930.

Ward, C., Gezari, S., Nugent, P et al. 'Panic at the ISCO: time-varying double-peaked broad lines from evolving accretion disks are common amongst optically variable AGN', **2024,** The Astrophysical Journal, 961, 172.

Ward, C., Gezari, S., Nugent, P et al. 'Variability-selected intermediate mass black hole candidates in dwarf galaxies from ZTF and WISE', **2022**, The Astrophysical Journal, 936, 104.

Ward, C., Gezari, S., Frederick, S. et al. 'AGNs on the Move: A Search for Off-nuclear AGNs from Recoiling Supermassive Black Holes and Ongoing Galaxy Mergers with the Zwicky Transient Facility', **2021**, The Astrophysical Journal, 913, 102.

Klare, C.*, **Ward, C.**, et al. 'Year-timescale changes in AGN radio luminosity as seen by the ASKAP Variables and Slow Transients Survey', submitted to ApJ, arXiv: 2608.15711.

Birmingham, S.†, **Ward, C.** et al. 'The birth of young radio jets in changing-look AGN: a population study', submitted to ApJ, arXiv: 2507.01355.

Gulanti, A.* et al (author 5 of 27), 'ASKAP J0055−2558: A Luminous, Long-Lived Radio Transient at z = 0.1 – an Orphan Afterglow or a Jetted TDE from an IMBH?', **2026**, ApJ, 1000, 118.

Melchior P., **Ward, C.** et al. 'scarlet2: Astronomical scene modeling in jax', **2026**, Journal of Open Source Software, 11(120), 9646, https://doi.org/10.21105/joss.09646.

Kaaz, N.$^{+}$, Liska, M., Ward, C. et al. 'Changing-look AGN Powered by Disk Tearing', ApJ, 2026, 1003, 161.

Stein, R.$^{+}$, Chornock, R., Ward, C. et al**.** 'TDE 2025abcr: A Tidal Disruption Event in the Outskirts of a Massive Galaxy', **2026,** ApJL, 1006, L57.

Townsend, A.* et al. (including Ward, C.) 'Follow-up of SN 2025wny IV: Photometric Time-delay Measurements of a Strongly Gravitationally Lensed Superluminous Supernova', submitted to MNRAS, arXiv:2608.28427.

Sfaradi, I.$^{+}$ et al. (including Ward, C.) 'An Off-Nuclear Tidal Disruption Event Discovered At Late Times: The Case Of TDE 2023mfm', submitted to ApJ.

Davis, G.† et al. (including Ward, C.) 'SN2025aico: An Interesting Case Of ^{56}Ni Mixing, Ejecta Asymmetries, and Dust Formation in a Type IIb Supernova, submitted to ApJ, arXiv: 2609.26935.

Ward, C., Melchior, P., et al. 'The Euclid-Rubin Derived Data Products: joint pixel processing for optimal galaxy and transient modeling.', in prep.

Zak, M*, **Ward, C.**, et al. 'Correlated Millimeter and X-Ray Variability in BASS AGN', in prep.

Sippy, K.*, **Ward, C.**, et al, 'Identification of non-nuclear massive black holes in the first data from Rubin and Euclid', in prep.

Krishnaraj, V.†, **Ward, C.**, et al. 'Strongly lensed supernovae in focus: Deblending marginally resolved systems in ground and space-based imaging', in prep.

Taggart, K.$^{+}$ et al. (including Ward, C.) 'Pre- and Post-Explosion JWST Observations of the Type Ibn Supernova 2023fyq Reveal Silicate Emission Features in a Pre-Existing Circumstellar Dust Reservoir', in prep.

Saad, S.† et al. (including Ward, C.), 'Reconsidering Tick Tock: An Alternative to the Supermassive Black Hole Binary Interpretation of J1430', in prep.

Yao, Y.$^{+}$, Chornock, R., **Ward, C.** et al. 'A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd', **2025**, ApJL, 985, L48.

Onoue, M. et al., (author 7 of 35), 'A Post-Starburst Pathway to Forming Massive Galaxies and Their Black Holes at z>6', **2025**, Nature Astronomy, doi:10.1038/s41550-025-02628-1.

Liu, Y.[†], Burke, C. et al. (author 7 of 7), 'Dwarf Active Galactic Nuclei from Variability for the Origins of Seeds (DAVOS): Properties of Variability-Selected AGNs in the Dark Energy Survey Deep Fields', **2025,** ApJ, 994, 162.

Miller et al., (including Ward, C.), 'The La Silla Schmidt Southern Survey', **2025**, PASP, 137, 9, doi:10.1088/1538-3873/ae02c5.

Dong et al., (including Ward, C.), 'Enabling Early Transient Discovery in LSST via Difference Imaging with DECam', **2025**, ApJL, 994, L8.

Chandler et al., (including Ward, C.), 'NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)', **2025**, arXiv:2507.13409.

Greenstreet et al., (including Ward, C.) 'Lightcurves, rotation periods, and colors for Vera C. Rubin Observatory's first asteroid discoveries', **2025**, in press ApJL.

Sampson, M.*, Melchior, P., **Ward, C.,** Birmingham, S.[†] 'Score matching diffusion models as data-driven priors for improved multi-band source separation', **2024**, Astronomy and Computing, Volume 49, id.100875, doi:10.1016/j.ascom.2024.100875.

Uppal, A.[†], **Ward C.,** et al. 'Astrometric Jitter as a Detection Diagnostic for Recoiling and Slingshot Supermassive Black Hole Candidates', **2024,** ApJ, 975, 286.

Ridley, E.*****, Nicholl, M., **Ward, C.** et al. 'AT2017bcc: time-varying double-peaked emission lines following the sudden ignition of a dormant galactic nucleus', **2024**, MNRAS, 531, 1905.

Burke, C.[+], Liu, Y.[†], **Ward, C.** et al. 'Dwarf AGNs from Variability for the Origins of Seeds (DAVOS): Properties of Variability-Selected AGNs in the COSMOS Field and Expectations for Rubin Observatory', **2024**, ApJ, 971, 140.

Liang, Y.*, Melchior, P., et al. (author 7 of 7) 'Outlier Detection in the DESI Bright Galaxy Survey', **2023**, ApJL, 956, L6.

Liang, Y.*, Melchior, P. et al. (author 5 of 5), 'Autoencoding Galaxy Spectra II: Redshift Invariance and Outlier Detection, **2023**, ApJ, 166, 75.

Hammerstein, E.*, Gezari, S. et al. (author 7 of 9), 'Integral Field Spectroscopy of 13 Tidal Disruption Event Hosts from the ZTF Survey', **2023**, ApJ, 957, 86.

Arcodia, R. et al (author 17 of 19), 'O Corona, where art thou? eROSITA's view of UV-optical-IR variability-selected massive black holes in low-mass galaxies', **2023**, A&A, 681, A97.

Brightman, M., **Ward, C.** et al. 'A Luminous X-Ray Transient in SDSS J143359.16+400636.0: A Likely Tidal Disruption Event', **2021**, The Astrophysical Journal, 909, 102.

Hammerstein, E.*, Gezari, S., van Velzen, S. et al. (author of 6 of 20) 'Tidal Disruption Event Hosts Are Green and Centrally Concentrated: Signatures of a Post-merger System', **2021**, ApJL, 908, L20.

van Velzen, S., Hammerstein, E.*, Gezari, S., et al. (author 6 of 44) 'Seventeen Tidal Disruption Events from the First Half of ZTF Survey Observations: Entering a New Era of Population Studies', **2021**, ApJ, 908, 4.

Frederick, S. et al. (author 8 of 20), 'A Family Tree of Optical Transients from Narrow-Line Seyfert 1 Galaxies', **2021**, ApJ, 920, 56.

Stein, R. et al. (including Ward, C.), 'A tidal disruption event coincident with a high-energy neutrino', **2021**, Nature Astronomy, 5, 510.

Coppejans, D. L. et al. (author 22 of 38), 'A Mildly Relativistic Outflow from the Energetic, Fast-rising Blue Optical Transient CSS161010 in a Dwarf Galaxy', **2020**, ApJL 895, L23.

Andreoni, I. et al. (author 16 of 52), 'GROWTH on S190814bv: Deep Synoptic Limits on the Optical/Near-infrared Counterpart to a Neutron Star-Black Hole Merger', **2020**, ApJ, 890, 131.

van Velzen, S. et al. (author 15 of 41), 'The First Tidal Disruption Flare in ZTF: From Photometric Selection to Multi-wavelength Characterization', **2019**, ApJ, 172, 198.

Mahabal, A. et al. (author 15 of 50), 'Machine Learning for the Zwicky Transient Facility', **2019**, PASP, 131, 997.

Frederick, S, J. et al. (author 15 of 20), 'A New Class of Changing-look LINERs', **2019**, The Astrophysical Journal, 883, 31.

Hung, T. et al. (author 15 of 28), 'Discovery of Highly Blueshifted Broad Balmer and Metastable Helium Absorption Lines in a Tidal Disruption Event', ApJ, **2019**, 879, 119.

Duev, D. et al. (author 11 of 11), Real-bogus classification for the Zwicky Transient Facility using deep learning', **2019**, MNRAS, 489, 3582.

van Roestal, J. et al. (author 10 of 12), 'Simultaneous Observations of the Northern TESS Sectors by the Zwicky Transient Facility', **2019**, RNAAS, 3, 9, 136.

Perley, D. et al. (author 63 of 65), 'The fast, luminous ultraviolet transient AT2018cow: extreme supernova, or disruption of a star by an intermediate-mass black hole?', **2019**, The Monthly Notices of the Royal Astronomical Society, 484, 1.

Dobie, D. . et al. (author 28 of 30), 'An ASKAP Search for a Radio Counterpart to the First High-significance Neutron Star—Black Hole Merger LIGO/Virgo S190814bv', **2019**, The Astrophysical Journal, 887, 13.

Nordin, J. et al. (including Ward, C.), 'Transient processing and analysis using AMPEL: alert management, photometry, and evaluation of light curves', **2019**, A&A, 631, 147.

Kerr, M. et al. (author 3 of 6), 'Extreme Scattering Events Towards Two Young Pulsars', **2018**, MNRAS, 474, 4.

Hobbs, G. et al. (author 9 of 78), 'A pilot ASKAP survey of radio transient events in the region around the intermittent pulsar PSR J1107- 5907', **2016**, MNRAS, 456, 4.

Lynch, C. et al. (author 6 of 6), 'Radio detections of southern ultracool dwarfs', **2016**, The Monthly Notices of the Royal Astronomical Society, 457, 2.

White Papers:

Ward, C., Nugent P. & Strauss M., 'After SDSS-V Blue Skies Ideas: Unveiling transients and their host galaxies in the era of time-domain surveys', internal white paper, **2024.**

Nyland, K. et al. (including **Ward, C.)** 'VLASS Epoch 4 Science Case' white paper, **2024,** https://science.nrao.edu/vlass/library/white-papers.

References

Prof. Peter Melchior
Assistant Professor, Department of Astrophysical Sciences, Princeton University.
melchior@astro.princeton.edu
Prof. Michael Strauss
Professor and Chair, Department of Astrophysical Sciences, Princeton University.
strauss@astro.princeton.edu
Prof. Suvi Gezari
Professor, Department of Astronomy, University of Maryland.
suvi@umd.edu
Dr. Peter Nugent
Senior Scientist, Dept. Head for Computational Science, Lawrence Berkeley National Laboratory.
PENugent@lbl.gov
Prof. Jenny Greene
Professor, Department of Astrophysical Sciences, Princeton University.
jgreene@astro.princeton.edu